Module 2: Kappa Bound (80-bit Long Double, C Compiler)

Battle Plan v1.6 -- David Fox -- May 2026

Claim

The kappa bound is computed using 80-bit long double arithmetic in C (gcc, -O3). The result kappa = 4.84330141977803 (approximately) is the certified output of the binary bin/print_kappa compiled from bin/print_kappa.c.

Source

```
bin/print_kappa.c | gcc -O3 -std=c11 | 80-bit long double
```

Certified Values

Item	Value
`kappa (80-bit long double)`	`4.8433014197780389`
`Compiler flags`	`gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm`
`Stdout SHA-256`	`3716c7dbb32524074b8fffb65eea4506...`
`Source SHA-256`	`bin/print_kappa.c`
`Arithmetic`	`80-bit extended precision (x86 long double, ~18-19 sig. digits)`

Role in Pipeline

The kappa bound (Module 2) establishes the irrationality measure exponent for pi/10, which feeds into the continued fraction analysis of Module 3 and validates the prime bound used in Module 4. The 80-bit long double provides approximately 18-19 significant decimal digits, sufficient for the required bound. The C binary is pre-compiled and its stdout is SHA-bound; recompile with the above flags if needed.